5.4 Solving Radical Equations and Inequalities

Name _____ Period ____ Date _____

Algebra II 5.4 (30-minute plan) · Student sheet

I can…

- solve equations that contain radicals or rational exponents.
- identify extraneous solutions.
- solve radical inequalities.

Vocabulary · fill in the missing word

Radical equation — an equation with a _____ in a radicand

Extraneous solution — an _____ solution that does not work in the original equation

Radical inequality — an inequality with a _____ in a radicand; the radicand must also be ≥ 0 for even roots

Part 1 · Notes and examples (fill in each note, then work the example)

Solving a radical equation: Step 1: _____ the radical. Step 2: raise both sides to the same _____ as the index. Step 3: solve. Step 4: _____ every answer in the _____ equation.

Example 1

Solve $2\sqrt{x} + 5 = 11$.

Extraneous solution: An **extraneous solution** is an answer that _____.

Example 2

Solve $x + 1 = \sqrt{7x + 15}$.

Rational exponents: To solve x^(m/n) = b, raise both sides to the _____ power.

Radical inequalities: For an inequality with an even root, the radicand must also be _____.

Example 3

Solve $3\sqrt{x - 1} \le 12$.

Part 2 · Partner practice

Partner practice

Solve $\sqrt{4x - 3} + 2 = 7$.